

Hammad Rizwan

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Halifax, Canada

EDUCATION

Dalhousie University Sept. 2023 –
PhD Computer Science, Advisor: Hassan Sajjad

Lahore University of Management Sciences (LUMS) Sept. 2018 – Jun 2020
MS Computer Science, Advisor: Asim Karim

RESEARCH INTERESTS

Model Editing, Machine Unlearning, Interpretability and Self-supervised Learning.

SKILLS

ML/AI: PyTorch, TensorFlow, HuggingFace, Transformers, Scikit-learn, LLM Fine-tuning, LoRA/Adapters, Self-Supervised Learning, Rasa

Programming: Python, C, Java, Go, Bash

Data & Analysis: NumPy, Pandas, Scipy, Matplotlib, Seaborn, Jupyter

Backend: FastAPI, Flask, REST API, Unicorn, Postman

Databases / Retrieval: Qdrant, FAISS, MongoDB, Redis, Elasticsearch

MLOps / Deployment: Docker, HPC/Slurm, Conda/Virtual Environments

Systems & Tools: Linux, Git, LaTeX

INDUSTRY EXPERIENCE

Upbeing.ai **Mitacs Accelerate Intern** Feb. 2024 – Nov. 2024

- Built and deployed a conversational AI system for a mental health support application, **combining large language model (LLM) based dialogue models, retrieval augmented generation (RAG), and representation similarity search pipelines** to deliver accurate, context-aware responses.
- Collaborated with cross-functional researchers and engineers to integrate **LLM fine-tuning and evaluation workflows**, improving system reliability and advancing **AI-driven mental health research**.

Arbisoft **Senior ML Engineer** Feb. 2023 – July 2023

- Built an end-to-end **personalized, machine learning-driven recommendation system** for a dining-focused social media platform, leveraging embedding models to **improve user engagement and content discovery**.
- Optimized entity extraction** on noisy short-text inputs, increasing downstream **real-time search relevance** and **recommendation quality**.

ISSM.ai **NLP Engineer** Oct. 2022 – Feb. 2023

- Built **enterprise conversational AI systems** (chatbots) for UBL and Allied Bank, applying **intent recognition, dialogue management, and information retrieval** techniques to improve query resolution time and enhance customer experience.
- Developed and deployed **multilingual NLP pipelines** for AI-driven automation of banking workflows in Urdu/Roman Urdu, **training domain-specific embedding LLMs** and leveraging transliteration techniques to reduce manual processing and improve operational efficiency.

Eighty.ai **NLP Engineer** Jan. 2021 – Dec. 2021

- Designed and deployed a **machine learning-driven HS code classification** pipeline, tackling a **large-scale, fine-grained prediction** task spanning 97 categories and 1,244 subcategories, and improving **accuracy and scalability** for **global trade and compliance workflows**.
- Trained **NER models and domain-specific embedding LLMs** to build a high-accuracy document processing pipeline for the financial domain, enabling **automated information extraction, entity normalization, and downstream compliance automation**.

- Developed a risk detection pipeline for **AIA Insurance**, leveraging **entity resolution and name-matching algorithms** to flag high-risk individuals and corporations for anti-money laundering (AML), improving the accuracy and reliability of compliance monitoring.
- Enhanced ComplyXPRT, a regulatory compliance platform for Barclays Bank, by integrating lifelong learning for adaptive modelling and achieving a **30% reduction in inference latency** through **model distillation, optimized deployment, and multi-threaded inference**.

RESEARCH EXPERIENCE

HyperMatrix Lab

Sept. 2023 –

Doctoral Researcher

- Investigating **knowledge editing in LLMs**, with a focus on lexical bias disentanglement and weight-preserving parameterization to enable controlled factual updates while minimizing collateral interference in unrelated knowledge.
- Designing **instance-level difficulty frameworks for machine unlearning**, leveraging sample-level hardness estimation.
- Contributing to **interpretability research**, including **range-based attribution methods** that characterize neurons over activation intervals, and ongoing work on **model pruning** for circuit discovery.

Robotics and Intelligent Computing (RICE) Lab

Nov. 2020 – June 2022

Research Associate, Digital Forestry

- Designed a **hybrid vision pipeline** for monocular tree diameter estimation, combining **deep learning-based segmentation with classical feature extraction and regression-based post-processing**, addressing challenges of unstructured outdoor forestry imagery.
- Created the first **annotated dataset for tree classification and semantic segmentation** in Pakistan, supporting downstream research and deployments by **WWF and Forestry Pakistan**.
- Contributed to scalable, low-cost computer vision systems for national-scale digital forestry initiatives.

Knowledge and Data Engineering (KADE) Lab

Sept. 2019 – June 2020

Research Assistant, Hate Speech and Generative Models

- Created the first **open-source datasets for offensive and hate speech detection in Roman Urdu**, and introduced a **custom CNN-ngram model** that established strong baselines; the work has since been **cited over 100 times** and serves as a benchmark in hate speech detection in low-resource NLP research.
- Explored **GAN-based paraphrase generation**, designing differentiable surrogate objectives to address the non-differentiability of discrete text outputs for the generator.

PUBLICATIONS

Peer-reviewed

- **Hammad Rizwan**, Domenic Rosati, Ga Wu, Hassan Sajjad “Resolving Lexical Bias in Model Editing” *Proceedings of the 42nd International Conference on Machine Learning (ICML), 2025*. [Paper Link] (26.9% acceptance rate)
- Ubaid Azam, **Hammad Rizwan**, Asim Karim “Exploring Data Augmentation Strategies for Hate Speech Detection in Roman Urdu.” *Proceedings of the Thirteenth Language Resources and Evaluation Conference (LREC), 2022*. [Paper Link]
- **Hammad Rizwan**, Muhammad Haroon Shakeel, Asim Karim “Hate-Speech and Offensive Language Detection in Roman Urdu.” *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP), 2020*. [Paper Link] (24.5% acceptance rate)

Preprints

- Muhammad Umair Haider, **Hammad Rizwan**, Hassan Sajjad, Peizhong Ju, AB Siddique “Neurons Speak in Ranges: Breaking Free from Discrete Neuronal Attribution”, 2025 (Under Review) [Paper Link]
- **Hammad Rizwan***, Mahtab Sarvmali*, Hassan Sajjad, Ga Wu “Instance-Level Difficulty: A Missing Perspective in Machine Unlearning”, 2025 [Paper Link]
- **Hammad Rizwan***, Ubaid Azam*, Faizad Ullah, Asim Karim “Towards Explainable Hate Speech Detection in Roman Urdu” [TBD Link]

AWARDS AND SCHOLARSHIPS

Doctoral Research Scholarship	2023 – 2027
Undergraduate Merit Scholarship (50% Tuition Coverage)	2014 – 2017

INVITED TALKS

Model Interpretability and Model Editing <i>Deep Learning for NLP – Dalhousie University</i>	2025
Exploration of Neuronal Activations <i>AI Cluster – Dalhousie University</i>	2024
Computer Vision for Digital Forestry <i>National Agriculture Robotics Workshop – LUMS</i>	2022

PROFESSIONAL SERVICES

- Co-Organiser: Machine Learning Theory Seminar - Dalhousie University (2024 -)
- Conference Reviewer: NeurIPS 2024 - ; ICML 2025 - ; ICLR 2025 -
- Journal Reviewer: LRE, IPM, NLE